MUSANZE HARVEST INTELLIGENCE SERVICE

SWE 3513 — Artificial Intelligence
Individual Assignment 2

Student: ABIJURU CYNTHIA
Registration number: 25/27096
Institutional username: ug2527096@ines.ac.rw
Verification code: AI-A2-7096
Operating system: Linux Ubuntu

Technical report generated from the reproducible pipeline. The dataset is synthetic and is intended only for coursework and demonstration.

1. Identity and decision problem

The stakeholder is a potato cooperative in Musanze. Farm records can arrive in inconsistent formats and internet access may be unreliable. The prototype addresses three questions: (1) what harvest quantity might be expected from the supplied field conditions, (2) should a record be flagged as high production risk, and (3) which records have similar input conditions?

The service is decision support only. It must not replace an agronomist or determine access to finance, insurance, or public services. Any prediction requires human review.

Individual reproducibility signature

Item	Value
Student	ABIJURU CYNTHIA
Registration number	25/27096
Seed (last four digits)	7096
Verification code	AI-A2-7096
Dataset SHA-256	de882fcc0a9bc61d1d42e6f37253fddceb1429214b63f9136a1b36261f605f0e

2. Data and reproducibility

The generator uses numpy.random.default_rng(7096). The base dataset contains 243 records because 240 + (7096 mod 41) = 243. It appends three exact duplicate rows, resulting in 246 raw rows. The CSV is written with four decimal places. The SHA-256 fingerprint is computed from the exact saved CSV bytes.

3. Data dictionary and cleaning

Column	Role / generation
rainfall_mm	Numeric feature; uniform 600–1400
avg_temp_c	Numeric feature; uniform 12–26
soil_moisture_pct	Numeric feature; uniform 20–80 before deliberate missingness
fertilizer_kg	Numeric feature; uniform 0–250 before deliberate missingness
field_size_ha	Numeric feature; uniform 0.2–4.0
season	Categorical feature; A or B
cooperative_support	Binary feature; 0 or 1
yield_tons	Regression target from assignment formula; clipped to minimum 0.50
high_risk	Classification target: yield per hectare below the dataset median

Data-quality evidence

Check	Observed result
Raw rows	246
Rows after exact duplicate removal	243
Duplicate rows removed	3
Missing soil moisture	12
Missing fertilizer	12

Missing values are not manually edited in the CSV. Numeric imputation and categorical imputation are fitted inside the preprocessing pipelines on training data for supervised tasks. Numeric features are scaled and season is one-hot encoded. Yield, yield per hectare and high_risk are excluded from input features to prevent target leakage.

4. Pipeline design and regression

The runner validates the seed against the final four digits in the supplied student identifier, regenerates the dataset, removes exact duplicates, validates observed ranges, and trains the three required task types. A fixed train/test split uses random_state=7096. Transformations are kept inside scikit-learn pipelines so training and inference use the same processing steps.

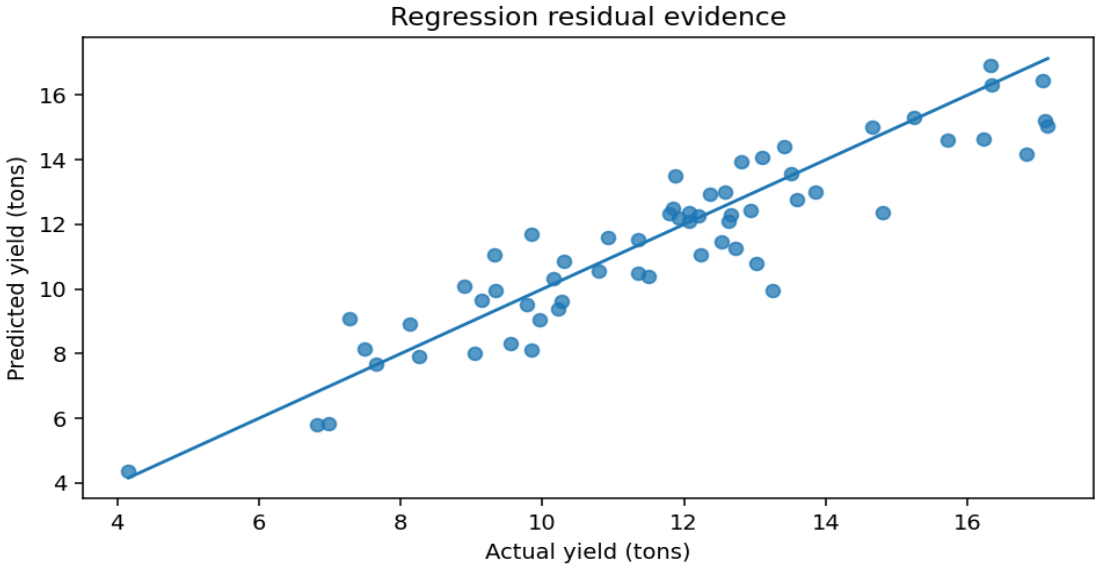
Regression metrics on the held-out test set

Model	MAE	RMSE	R^2
mean_baseline	2.2705	2.8501	-0.0015
linear_regression	0.9130	1.1528	0.8362
random_forest	1.1170	1.4661	0.7350

Selected model: **linear_regression**. It had the lower held-out RMSE among the two candidate regressors. Its MAE was 0.913 tons, meaning the average absolute prediction error was about that many tons on this synthetic test split. The mean baseline was substantially weaker, while linear regression outperformed the random forest for this generated formula.

Regression error interpretation

The MAE and RMSE are expressed in tons. RMSE penalizes larger errors more strongly than MAE. These values are internal synthetic-data measurements; they do not establish accuracy on real farms. Review residuals.png to see the relationship between actual and predicted test yields.



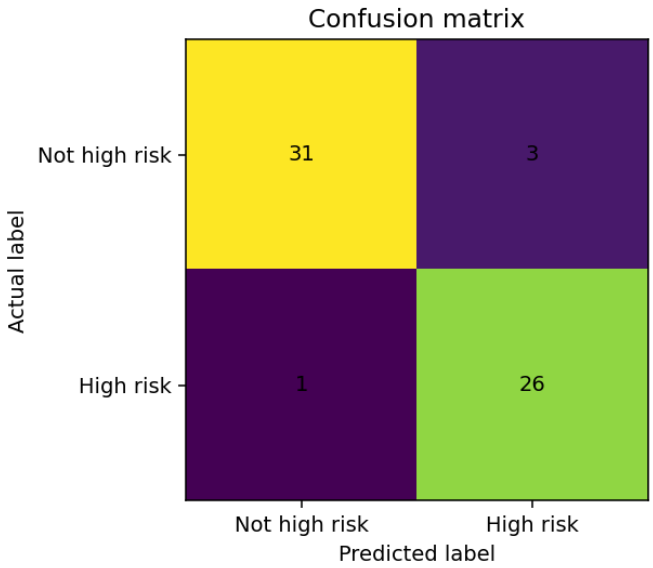
5. Classification

The target high_risk equals 1 when generated yield per hectare is below the median. Classification input features include only the seven field-condition features. The target and derived yield measures are not predictors.

Model	Precision	Recall	F1	ROC-AUC
majority_baseline	0.443	1.000	0.614	0.500
logistic_regression	0.897	0.963	0.929	0.985
random_forest	0.862	0.926	0.893	0.965

Selected model: **logistic_regression**, chosen by the highest held-out F1 among candidate classifiers. Its confusion matrix is [[31, 3], [1, 26]] in the order actual class 0/1 by predicted class 0/1. This means 31 class-0 records and 26 high-risk records were correctly classified, with three false positives and one false negative.

The false negative is especially important because a high-risk case was predicted as not high-risk. In real use, staff should review uncertain records and not treat this output as a final agronomic judgment. ROC-AUC is reported because both classes occur in the test split.



6. Clustering and profiles

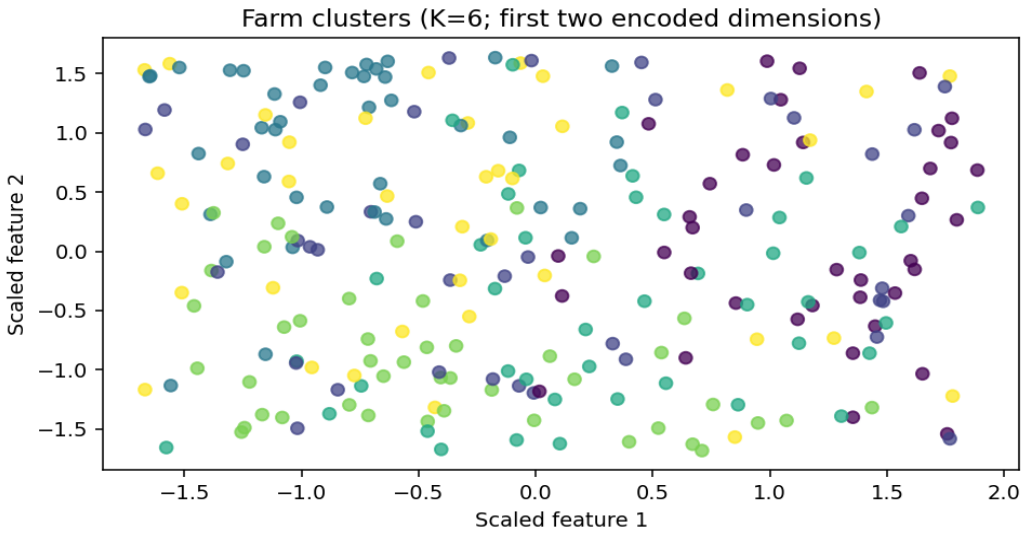
K-means clustering was tested for K=2 through K=6 after imputation, encoding and scaling of input features. Target columns are not included. The chosen K is the one with the highest measured silhouette score in this range.

K	Silhouette	Inertia
2	0.1570	1333.35
3	0.1273	1224.02
4	0.1456	1115.70
5	0.1475	1024.39
6	0.1577	945.84

Selected K = 6. Its silhouette score is 0.1577. This is the highest of the tested values, although the score is modest, so separation between clusters is limited.

Cluster	Records	Rainfall mean	Moisture mean	Fertilizer mean	Field size mean
0	38	1231.3	48.5	124.2	2.51
1	39	987.6	43.7	74.8	3.18
2	41	800.5	40.4	125.2	1.92
3	42	1044.3	45.8	102.3	1.09
4	45	885.0	61.1	107.6	1.65
5	38	904.5	67.8	181.1	2.47

Operational interpretation: profiles summarize which generated field conditions are similar. A cluster is not itself a risk category, and clusters should be re-evaluated with representative real farm data before operational use.



7. API, tests and responsible AI

The FastAPI service provides GET /health, GET /model-info, and POST /predict. Pydantic validates types and required ranges; extra fields are rejected. The CLI loads the same saved model artifacts and accepts one JSON record. Invalid input returns a useful validation error rather than a prediction.

Evidence	Result
Automated tests	9 tests passed during preparation
Valid prediction	samples/valid.json
Invalid prediction	samples/invalid.json (rainfall exceeds allowed maximum)
API docs	http://127.0.0.1:8000/docs after starting uvicorn
Saved models	models/regression.joblib; classification.joblib; clustering.joblib

Limitations and risks

All records are synthetic and the target is created by a known formula with random noise. Metrics may therefore look better than results on real farm data. Real-world data can have different distributions, measurement errors, missingness patterns, and relationships. Before deployment, obtain representative data with appropriate permissions, validate performance externally, monitor model drift, document data provenance, and obtain agronomist review. The prototype must not determine access to credit, insurance, or public services.

Reproduction commands

```
python run_all.py --student-id ug2527096@ines.ac.rw --seed 7096
python predict.py --json samples/valid.json
pytest -q
uvicorn app:app --host 127.0.0.1 --port 8000
```

Submission note: the student must record the required unedited demonstration video personally, verify the generated files on their Ubuntu machine, add the real repository URL and commit hash, and complete/sign the declaration truthfully.